(2) (b)

The proportional odds model as defined in paper is of the form

$$k_j(x) = k_j \exp(-\beta^T x)$$

or $\log\left(\frac{\gamma_j(x)}{1-\gamma_j(x)}\right) = \theta_j - \beta^T X \quad (1 \le j \le k)$

where $\frac{\gamma_j(x)}{1-\gamma_j(x)}$ is the odds ratio for the event.

$$\gamma_j(x) = \pi_1(x) + \pi_2(x) \cdots \pi_j(x)$$

i.e $= \sum_{h=1}^{j} \pi_h(x)$

multinomial observation $(n_1\ n_2 \cdots n_k)$ to the likelihood function is $\pi_1^{n_1}, \pi_2^{n_2} \cdots \pi_k^{n_k}$, with probabilities satisfying

$$\text{link}(\gamma_{ij}) = (\theta_j - \beta^T X_i) / \tau_i$$

For the cummulative probabilities in ordinal regression, we define,

$R_1 = n_1$ $\qquad$ $Z_1 = R_1/n$

$R_2 = n_1 + n_2$ $\qquad$ $Z_2 = R_2/n$

$\vdots$ $\qquad$ $\vdots$

$R_K = \sum n_j = n$ $\qquad$ $Z_K = R_K/n$

$\Rightarrow$ Now for Maximum likelihood estimate we find $P(X/\theta)$

i.e $\underset{w}{\operatorname{argmax}}\, P(X/\theta) = \prod_{i=1}^{K} P(X_i/\theta)$

$P(X_i/\theta) = P(R_i : R_{i+1} - R_i \mid R_{i+1})$

i.e probability that R_i gets divided in the ratio R_i & $R_{i+1} - R_i$

$$= \left(\frac{\gamma_i}{\gamma_{i+1}}\right)^{R_i} \left(\frac{\gamma_{i+1} - \gamma_i}{\gamma_{i+1}}\right)^{R_{i+1} - R_i}$$

i.e $$P(X_i/\theta) = \prod_{i=1}^{K} \left\{ \left(\frac{\gamma_i}{\gamma_{i+1}}\right)^{R_i} \left(\frac{\gamma_{i+1} - \gamma_i}{\gamma_{i+1}}\right)^{R_{i+1} - R_i} \right\}$$

taking log on both sides

$$\underset{\theta}{\operatorname{argmax}}\ L(\theta)$$

$$= \sum_{i=1}^{k} \left\{ R_i \operatorname{Log}\left(\frac{\gamma_i}{\gamma_{i+1}}\right) + (R_{i+1} - R_i) \log\left(\frac{\gamma_{i+1} - \gamma_i}{\gamma_{i+1}}\right) \right\}$$

$$= \sum_{i=1}^{k} \left\{ R_i \operatorname{Log}\left(\frac{\gamma_i}{\gamma_{i+1} - \gamma_i}\right) - R_{i+1} \operatorname{Log}\left(\frac{\gamma_{i+1}}{\gamma_{i+1} - \gamma_i}\right) \right\}$$

$$= n \sum_{i=1}^{k} \left\{ Z_i \varphi_i - Z_{i+1}\, g(\varphi_i) \right\} \rightarrow (2)$$

where $Z_i = R_i / n$

$$\varphi_i = \log\left(\frac{\gamma_i}{\gamma_{i+1} - \gamma_i}\right)$$

$$g(\varphi_i) = \log\left(\frac{\gamma_{i+1}}{\gamma_{i+1} - \gamma_i}\right)$$

now putting

$$E\left[\frac{\partial L}{\partial \varphi}\right] = 0$$

we get, $E\left[\frac{Z_i}{Z_{i+1}}\right] = Z_{i+1}\, g'(\varphi_i)$

$$= \frac{Z_{i+1}}{\gamma_{i+1}} * \gamma_i$$

$\Rightarrow E[Z_i] = \gamma_i$

Now as $Var(Z_i) = -\frac{\partial L}{\partial \theta^2}$ ②

$\Rightarrow$ therefor differentiating equation few times we get

$$Var\left(\frac{Z_i}{Z_{i+1}}\right) = \frac{Z_{i+1}\, g''(\theta_i)}{n}$$

$$\Rightarrow Var(Z_i) = \frac{\gamma_i (1-\gamma_i)}{n}$$